

Q1. Which of the following is the lightest and most abundant chemical element in the universe?

- A. Hydrogen
- B. Oxygen
- C. Nitrogen
- D. Carbon

Answer: Option 1

Solution: Hydrogen is the most abundant chemical element, making up about 75% of the universe's elemental mass. Hence, Option (1) is the right answer.

Q2. In the context of the kinetic theory of gases, which of the following properties is NOT assumed to be a characteristic of an ideal gas?

- A. Particles do not exert forces on each other except during collisions.
- B. Particles occupy negligible space.
- C. Particles have varying masses.
- D. All collisions are perfectly elastic.

Answer: Option 4

Solution: The kinetic theory assumes that all particles of an ideal gas have identical mass. The statement that particles have varying masses is incorrect. Hence, Option (4) is the right answer.

Q3. If a data set has values 3, 7, 7, 2, 5, 7, what is the mode of this data set?

- A. 7
- B. 5
- C. 6
- D. 3

Answer: Option 1

Solution: The mode is the number that appears most frequently in a data set. Here, the number 7 appears most frequently. Hence, Option (1) is the right answer.